

Algebraic Clock Drift and a Parity–Alias Replacement Obstruction

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Abstract

The exact-rational comparator of RH-329 freezes the exponential clock $\sigma_k = \widehat{\Lambda}^{-2k}$, where $\widehat{\Lambda} = 2098216888035403/1250000000000000$. We prove that this clock is not the physical fixed-phase clock. The physical expansion rate is the unique positive root of $p(x) = x^3 + 4x^2 - 16$, and exact rational arithmetic gives $p(\widehat{\Lambda}) > 0$; hence $\widehat{\Lambda} > \lambda$ and the physical phase tends to $-\infty$. Directly from the uniform parity remainder and the physical multiplier law, we derive an off-phase trichotomy valid for every clock $\sigma_k = \Lambda_c^{-2k}$ with $\Lambda_c > 1$. On the RH-329 clock the physical alias packet dominates both the physical parity packet and the two hatted model packets exponentially. The resulting scalar parity–alias comparator defect divided by the physical alias packet tends to -1 , and divided by the target $H_k = kR^{-2k}$ tends to $-\infty$. This rejects RH-329 only as a physical bounded-phase parity–alias comparator. It is not an actual full-trace divergence theorem and does not activate the fixed-phase transfer theorem of RH-330. Replacing the clock by the exact physical λ leaves target-scale replacement not testable: the archived relative $o(1)$ laws do not supply the exponentially sharper remainder $o((\beta R)^{-2k})$. No determinant gate or Riemann-hypothesis conclusion is obtained.

1 Physical and hatted clocks

Let $u_c \in (1, 2)$ be the physical parameter and put

$$\lambda = 2u_c(u_c - 1) > 1. \quad (1)$$

The algebraic identities of the gauge-fixed physical observation source give

$$p(\lambda) = 0, \quad p(x) = x^3 + 4x^2 - 16 \quad (2)$$

[4]. The RH-329 model instead defines the exact rational constant

$$\widehat{\Lambda} = \frac{2098216888035403}{1250000000000000} \quad (3)$$

and freezes $\sigma_k = \widehat{\Lambda}^{-2k}$ for integers $k \geq 2$ [2]. The hat is part of the model type. It is not an interval enclosure for the physical root.

For an arbitrary clock base $\Lambda_c > 1$, define

$$\sigma_k = \Lambda_c^{-2k}, \quad \eta_k = k - \frac{\log(1/\sigma_k)}{2 \log \lambda} = k \left(1 - \frac{\log \Lambda_c}{\log \lambda} \right). \quad (4)$$

Only $\Lambda_c = \lambda$ has bounded phase. This observation must be proved exactly for a proposed algebraic comparator; closeness of printed decimals is irrelevant.

Proof. On the stated clock,

$$\delta_{\sigma_k} = C_* \Lambda_c^{-k} \{1 + o(1)\}, \quad k\delta_{\sigma_k} \longrightarrow 0.$$

The uniform parity remainder from RH-326 is

$$\left| 1 - (1 - \delta)^{2k} - 2k\delta \right| \leq k(2k - 1)\delta^2.$$

After division by $2k\delta$, its right side is $O(k\delta) = o(1)$, proving (13). The multiplier law gives

$$\beta_k^{2k} = \frac{\beta^{2k}}{C_M} \{1 + o(1)\}.$$

Since $\beta^{2k} = r_H^{-2k} \lambda^{-k}$, the $2\beta^{2k}$ endpoint term is lower by a factor $O(k^{-1})$, and (14) follows. Division proves (15); the three cases follow from the strict ordering of positive clock bases. $\square$

3 Failure of the hatted fixed-phase comparator

The RH-329 comparator defines the exact model constants

$$\hat{C}_M = \frac{9731714526004839}{5000000000000000}, \quad \hat{C}_* = \frac{26314633984227}{2500000000000000} \quad (16)$$

and

$$\hat{\beta}^2 = \frac{1}{r_H^2 \hat{\Lambda}}, \quad \hat{\mathcal{A}}_k = \left(\frac{2k - 2}{\hat{C}_M} + 2 \right) \hat{\beta}^{2k}, \quad \hat{\mathcal{P}}_k = r_H^{-2k} \left\{ 1 - (1 - \hat{C}_* \hat{\Lambda}^{-k})^{2k} \right\}. \quad (17)$$

Again, these are model definitions, not physical interval data. Compare only the parity-alias scalar, with the RH-330 actual-minus-model sign convention:

$$D_k = (\mathcal{P}_k^{\text{route}} - \hat{\mathcal{P}}_k) - (\mathcal{A}_k^{\text{route}} - \hat{\mathcal{A}}_k), \quad \sigma_k = \hat{\Lambda}^{-2k}. \quad (18)$$

Theorem 3.1 (RH-329 parity-alias comparator obstruction). *On the RH-329 clock,*

$$\frac{D_k}{\mathcal{A}_k^{\text{route}}} \longrightarrow -1. \quad (19)$$

Moreover,

$$\frac{\mathcal{A}_k^{\text{route}}}{H_k} \longrightarrow +\infty, \quad \frac{D_k}{H_k} \longrightarrow -\infty. \quad (20)$$

Thus RH-329 is not a physical bounded-phase parity-alias comparator.

Proof. By theorem 2.1 and $\hat{\Lambda} > \lambda$,

$$\mathcal{P}_k^{\text{route}} = O\left(k r_H^{-2k} \hat{\Lambda}^{-k}\right) = o(\mathcal{A}_k^{\text{route}}).$$

The same uniform binomial upper bound and the exact formulas (17) give

$$\hat{\mathcal{P}}_k + \hat{\mathcal{A}}_k = O\left(k r_H^{-2k} \hat{\Lambda}^{-k}\right) = o(\mathcal{A}_k^{\text{route}}).$$

Expanding (18) therefore leaves $D_k = -\mathcal{A}_k^{\text{route}} \{1 + o(1)\}$, proving (19).

Next, (14) and (12) give

$$\frac{\mathcal{A}_k^{\text{route}}}{H_k} = \frac{2}{C_M} \left(\frac{R^2}{r_H^2 \lambda} \right)^k \{1 + o(1)\} = \frac{2}{C_M} (\beta R)^{2k} \{1 + o(1)\}. \quad (21)$$

The exact certificate $\beta R > 1$ is established in RH-336 [5], so the first limit in (20) follows. Multiplying it by (19) proves the second. $\square$

Remark 3.2 (Strict scope of the defect). D_k contains only the parity and alias replacement defects. It is not the complete signed replacement aggregate $\Delta\mathcal{B} + \Delta\mathcal{S} + \Delta\mathcal{R} + \Delta\mathcal{P} - \Delta\mathcal{A}$, not the actual five-slot coefficient, and not a full-trace residual. The boundary, shell, and far slots could still contribute at the same or larger scale. In addition, the sharp transfer theorem of RH-330 assumes one common bounded-phase clock [7]; theorem 1.1 proves that hypothesis false for the RH-329 clock. We therefore do not use that theorem to promote (20).

4 Correct-clock target-resolution barrier

It is natural to replace $\hat{\Lambda}$ by the exact physical λ . On that corrected clock, theorem 2.1 removes the exponential parity–alias mismatch, but it does not prove target-scale replacement.

Proposition 4.1 (Available asymptotics do not test the target remainder). *On the correct clock $\sigma_k = \lambda^{-2k}$,*

$$\frac{H_k}{\mathcal{A}_k^{\text{route}}} = \frac{C_M}{2}(\beta R)^{-2k}\{1 + o(1)\}. \quad (22)$$

Consequently a relative parity–alias replacement estimate sufficient for an $o(H_k)$ scalar defect must be

$$o\left(\frac{H_k}{\mathcal{A}_k^{\text{route}}}\right) = o\left((\beta R)^{-2k}\right). \quad (23)$$

*The archived laws (9) provide only relative $o(1)$ remainders. They neither imply nor refute (23); the correct-clock physical replacement is therefore **NOT-TESTABLE** from the present inputs.*

Proof. Equation (22) is the reciprocal of (21). Since $\beta R > 1$, the right side tends to zero exponentially. A bare relative $o(1)$ remainder has no specified rate and does not imply an exponentially weighted little-oh estimate. No lower bound in the archived source proves that such an estimate is impossible, so only the stated insufficiency verdict is justified. $\square$

This barrier is the same target-resolution issue isolated for the physical shell in RH-327 and the matching equation in RH-328 [3, 6]. It must not be replaced by a finite fit of C_* , C_M , or λ .

5 Reproduction protocol and claim boundary

The executable artifact evaluates the polynomial residual, hatted constants, finite model packets, and binomial enclosures with exact rational arithmetic. It separately prints high-precision clock diagnostics. The finite rows $k \in \{2, 3, 4, 6\}$ reproduce exact formulas only; they do not certify any moving-order physical asymptotic. The tests lock the exact root ordering, model types, packet bounds, sign convention, target barrier, and the complete claim firewall.

The maximum conclusion is scoped: the RH-329 hatted clock is off the physical bounded-phase route, and its scalar parity–alias comparison has the limits in theorem 3.1. No result here estimates the physical boundary/shell aggregate or signed far remainder, proves off-alias convergence, transports the noisy head, glues a determinant, or proves actual full-trace divergence. No Hilbert–Polya operator is constructed, no Riemann zero or von Mangoldt trace is identified, no completed-zeta divisor equality is proved, and no implication for the Riemann hypothesis is asserted. Gates A–E remain false/open.

References

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